Run Debug Stop Share Save { } Beautify

main.py

```python
def binary_search(arr, low, high, key):
    if low <= high:
        mid = (low + high) // 2

        if arr[mid] == key:
            return mid
        elif arr[mid] > key:
            return binary_search(arr, low, mid - 1, key)
        else:
            return binary_search(arr, mid + 1, high, key)

    return -1

arr = [5, 10, 15, 20, 25]
key = 20

result = binary_search(arr, 0, len(arr)-1, key)

if result != -1:
    print("Key found at index", result)
else:
    print("Key not found")
```

input

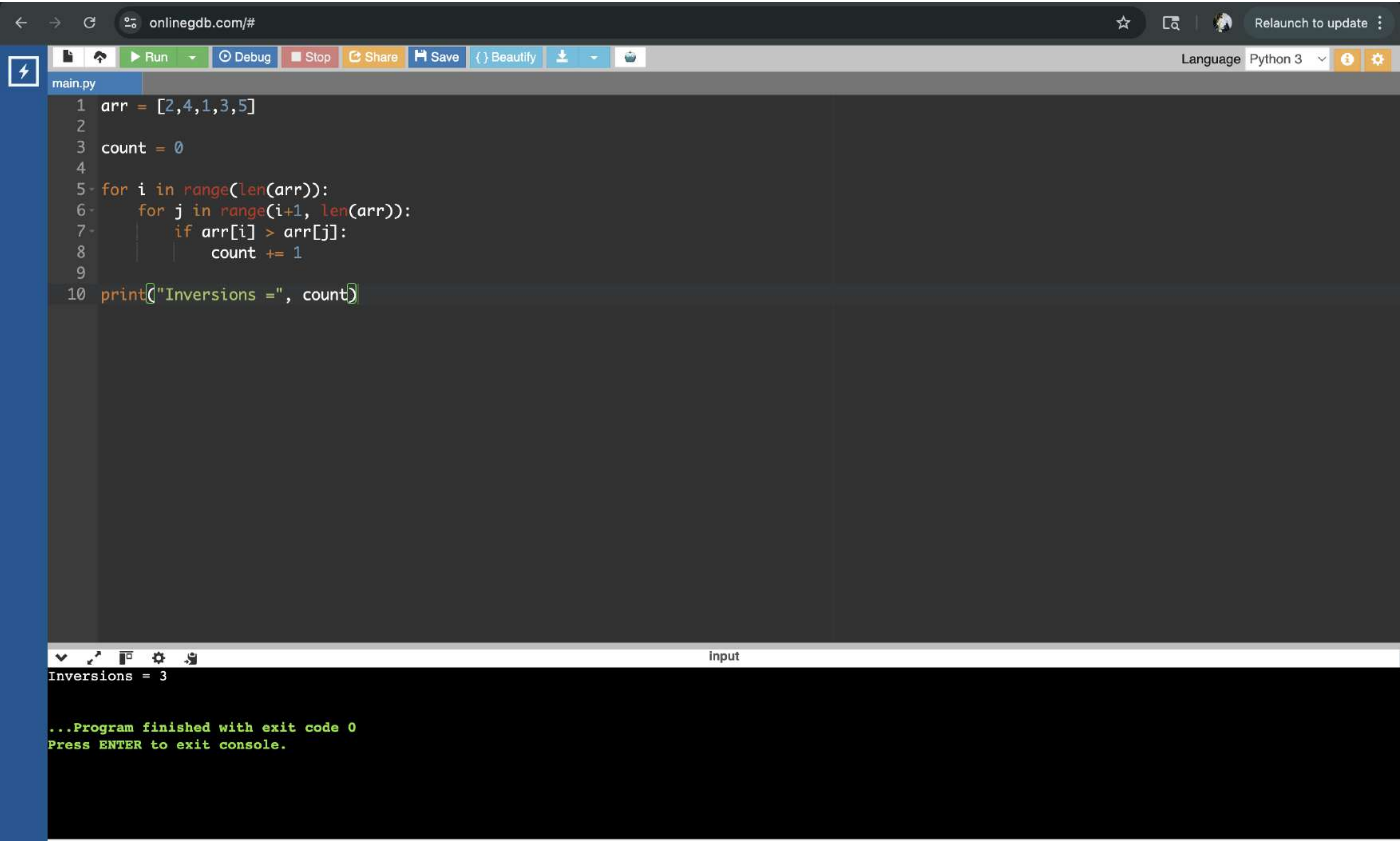
```
Key found at index 3


...Program finished with exit code 0
Press ENTER to exit console.
```

main.py

```python
def power(x, n):
    if n == 0:
        return 1

    half = power(x, n // 2)

    if n % 2 == 0:
        return half * half
    else:
        return x * half * half

print(power(2,10))
```

input

```
1024


...Program finished with exit code 0
Press ENTER to exit console.
```



onlinegdb.com/#
Relaunch to update
Run
Debug
Stop
Share
Save
{ } Beautify
Language
Python 3
main.py
arr = [2,4,1,3,5]
count = 0
for i in range(len(arr)):
for j in range(i+1, len(arr)):
if arr[i] > arr[j]:
count += 1
print("Inversions =", count)
input
Inversions = 3
...Program finished with exit code 0
Press ENTER to exit console.

main.py

```python
def power(x, n):
    if n == 0:
        return 1

    half = power(x, n // 2)

    if n % 2 == 0:
        return half * half
    else:
        return x * half * half

print(power(2,10))
```

input

```
1024


...Program finished with exit code 0
Press ENTER to exit console.
```

Run | Debug | Stop | Share | Save | { } Beautify

main.py

```python
# Iterative
n = 5
fact = 1

for i in range(1, n + 1):
    fact = fact * i

print("Iterative Factorial =", fact)

# Recursive
def factorial(n):
    if n == 0 or n == 1:
        return 1
    return n * factorial(n - 1)

print("Recursive Factorial =", factorial(5))
```

input

```
Iterative Factorial = 120
Recursive Factorial = 120


...Program finished with exit code 0
Press ENTER to exit console.
```

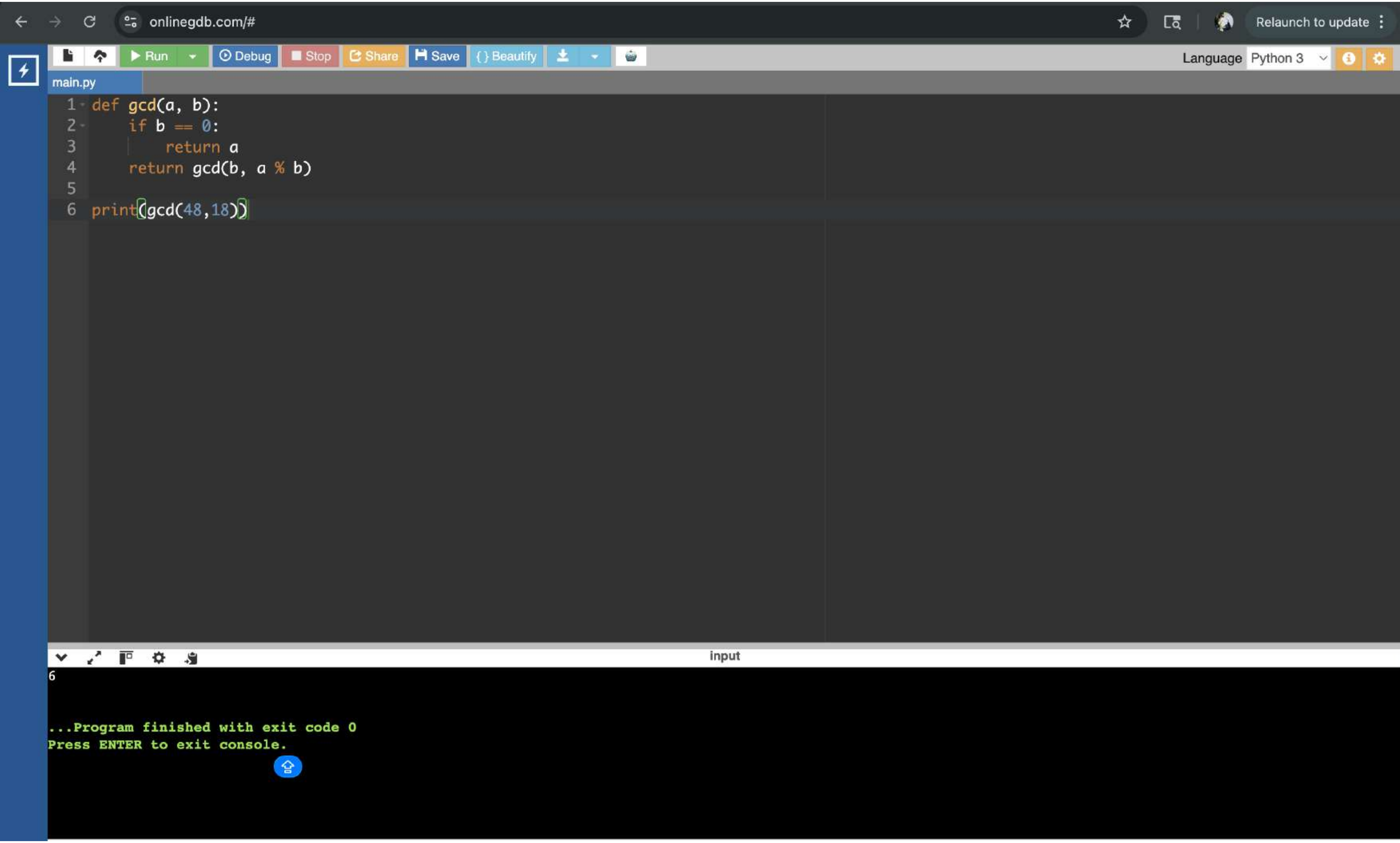
Run | Debug | Stop | Share | Save | { } Beautify | Language Python 3

main.py

```python
n = 6
a = 0
b = 1

for i in range(n):
    print(a, end=" ")
    c = a + b
    a = b
    b = c
```

input

```
0 1 1 2 3 5

...Program finished with exit code 0
Press ENTER to exit console.
```


onlinegdb.com/#
Relaunch to update
Run
Debug
Stop
Share
Save
{ } Beautify
Language
Python 3
main.py
def gcd(a, b):
    if b == 0:
        return a
    return gcd(b, a % b)

print(gcd(48,18))
input
6
...Program finished with exit code 0
Press ENTER to exit console.

main.py

```python
import heapq

arr = [4,10,3,5,1]

heapq.heapify(arr)

sorted_arr = []

while arr:
    sorted_arr.append(heapq.heappop(arr))

print(sorted_arr)
```

```
[1, 3, 4, 5, 10]


...Program finished with exit code 0
Press ENTER to exit console.
```

onlinegdb.com/#

main.py

```python
arr = [12,11,13,5,6]

for i in range(1, len(arr)):
    key = arr[i]
    j = i - 1

    while j >= 0 and arr[j] > key:
        arr[j + 1] = arr[j]
        j -= 1

    arr[j + 1] = key

print(arr)
```

input

```
[5, 6, 11, 12, 13]


...Program finished with exit code 0
Press ENTER to exit console.
```

Run Debug Stop Share Save { } Beautify

main.py

```python
arr = [10, 25, 30, 45, 50]
key = 30

found = False

for i in range(len(arr)):
    if arr[i] == key:
        print("Key found at index", i)
        found = True
        break

if not found:
    print("Key not found")
```

input

```
Key found at index 2


...Program finished with exit code 0
Press ENTER to exit console.
```

main.py

```python
A = [[1, 2],
     [3, 4]]

B = [[5, 6],
     [7, 8]]

C = [[0, 0],
     [0, 0]]

for i in range(2):
    for j in range(2):
        for k in range(2):
            C[i][j] += A[i][k] * B[k][j]

print(C)
```

input

```
0 1 1 2 3 5

...Program finished with exit code 0
Press ENTER to exit console.
```

main.py

```python
arr = [-2,-3,4,-1,-2,1,5,-3]

max_sum = arr[0]
current = arr[0]

for i in range(1, len(arr)):
    current = max(arr[i], current + arr[i])
    max_sum = max(max_sum, current)

print(max_sum)
```

```
7


...Program finished with exit code 0
Press ENTER to exit console.
```

Python Algorithm Experiment | Lab programs Topic 1 & 2 - C | GDB online Debugger | Comp

Ask Gemini

onlinegdb.com/#

Relaunch to update

Run | Debug | Stop | Share | Save | { } Beautify

Language Python 3

main.py

```python
def merge_sort(arr):
    if len(arr) > 1:

        mid = len(arr) // 2
        left = arr[:mid]
        right = arr[mid:]

        merge_sort(left)
        merge_sort(right)

        i = j = k = 0

        while i < len(left) and j < len(right):
            if left[i] < right[j]:
                arr[k] = left[i]
                i += 1
            else:
                arr[k] = right[j]
                j += 1
            k += 1

        while i < len(left):
            arr[k] = left[i]
            i += 1
            k += 1

```

input

```
0 1 1 2 3 5

...Program finished with exit code 0
Press ENTER to exit console.
```

main.py

```python
def quick_sort(arr):
    if len(arr) <= 1:
        return arr

    pivot = arr[0]

    left = []
    right = []

    for x in arr[1:]:
        if x <= pivot:
            left.append(x)
        else:
            right.append(x)

    return quick_sort(left) + [pivot] + quick_sort(right)

arr = [10,7,8,9,1,5]

print(quick_sort(arr))
```

```
[1, 5, 7, 8, 9, 10]


...Program finished with exit code 0
Press ENTER to exit console.
```

Run | Debug | Stop | Share | Save | { } Beautify | Language Python 3

main.py

```python
def hanoi(n, source, helper, destination):
    if n == 1:
        print(source, "->", destination)
        return

    hanoi(n-1, source, destination, helper)
    print(source, "->", destination)
    hanoi(n-1, helper, source, destination)

hanoi(3, "A", "B", "C")
```

input

```
A -> B
C -> B
A -> C
B -> A
B -> C
A -> C


...Program finished with exit code 0
Press ENTER to exit console.
```